ClipSAT : ACT International Subject Test: Mathematics 2 Level 2

Unit 1.01: Complex Numbers & De Moivre Theorem

Name: _____

Q1. For the complex number i = sqrt{-1}, which of the following is equivalent to the expression 5 - 3i - 2 - 7i?

[ ] A: 3 + 4i
[ ] B: 3 - 10i
[ ] C: 7 + 4i
[ ] D: 3 - 4i

Q2. For what real value of k is the complex expression frac{k + 8i}{3 - i} + frac{2 - ki}{3 + i} a real number?

[ ] A: 11
[ ] B: 1
[ ] C: 6
[ ] D: -11

Q3. Let z be a complex number satisfying the equation |z - 12| + |z + 12| = 26. What is the maximum possible value of |z - 5i| ?

[ ] A: frac{169}{12}
[ ] B: sqrt{194}
[ ] C: 18
[ ] D: 10

Q4. For the imaginary unit i such that i^2 = -1 , what is the product 4 - 3i2 + 5i ?

[ ] A: 23 + 14i
[ ] B: -7 + 14i
[ ] C: 23 - 14i
[ ] D: 8 - 15i

Q5. For the imaginary unit i, where i^2 = -1, the expression frac{k + 6i}{3 - 2i} is equal to a real number for some real constant k. What is the value of k?

[ ] A: -9
[ ] B: -4
[ ] C: 4
[ ] D: 9

Q6. In the complex plane, the complex numbers z and w satisfy |z| = 5, |w| = 7, and |z + w| = 10. What is the value of |z - w|?

[ ] A: 4sqrt{3}
[ ] B: 2sqrt{6}
[ ] C: 4sqrt{6}
[ ] D: 2sqrt{37}

Q7. For the imaginary unit i, where i^2 = -1, which of the following complex numbers is equivalent to 4 + 3i2 - 5i?

[ ] A: 23 - 14i
[ ] B: 23 + 14i
[ ] C: -7 - 14i
[ ] D: -7 + 14i

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Q8. In the complex number system, let i^2 = -1. For what real value of k is the expression frac{6 + ki}{3 + 2i} a purely imaginary number?

[ ] A: -9

[ ] B: -4

[ ] C: 4

[ ] D: 9